

IMPORTANT TOPIC

PAXOX Algorithm

<https://www.geeksforgeeks.org/operating-systems/paxos-algorithm-in-distributed-system/>

Byzantine Algorithm

<https://www.geeksforgeeks.org/operating-systems/paxos-algorithm-in-distributed-system/>

Consensus Mechanism

Proof of work

1. **Proof of Work (PoW):** This consensus algorithm is used to select a miner for the next block generation. Bitcoin uses this PoW consensus algorithm. The central idea behind this algorithm is to solve a complex mathematical puzzle and easily give out a solution. This mathematical puzzle requires a lot of computational power and thus, the node who solves the puzzle as soon as possible gets to mine the next block. For more details on PoW, please read [Proof of Work \(PoW\) Consensus](#)
2. **Practical Byzantine Fault Tolerance (PBFT):** Please refer to the existing article on [practical Byzantine Fault Tolerance \(PBFT\)](#).
3. **Proof of Stake (PoS):** This is the most common alternative to PoW. Ethereum has shifted from PoW to PoS consensus. In this type of consensus algorithm, instead of investing in expensive hardware to solve a complex puzzle, validators invest in the coins of the system by locking up some of their coins as stakes. After that, all the validators will start validating the blocks. Validators will validate blocks by placing a bet on them if they discover a block that they think can be added to the chain. Based on the actual blocks added in the Blockchain, all the validators get a reward proportionate to their bets, and their stake increase accordingly. In the end, a validator is chosen to generate a new block based on its economic stake in the network. Thus, PoS encourages validators through an incentive mechanism to reach to an agreement.
4. **Proof of Burn (PoB):** With PoB, instead of investing in expensive hardware equipment, validators 'burn' coins by sending them to an address from where they are irretrievable. By committing the coins to an unreachable address, validators earn the privilege to mine on the system based on a random selection process. Thus, burning coins here means that validators have a long-term commitment in exchange for their short-term loss. Depending on how the PoB is implemented, miners may burn the native currency of the Blockchain application or the currency of an alternative chain, such as bitcoin. The more coins they burn, the better their chances of being selected to mine the next block.

While PoB is an interesting alternative to PoW, the protocol still wastes resources needlessly. And it is also questioned that mining power simply goes to those who are willing to burn more money.

5. **Proof of Capacity:** In the Proof of Capacity consensus, validators are supposed to invest their hard drive space instead of investing in expensive hardware or burning coins. The more hard drive space validators have, the better their chances of getting selected for mining the next block and earning the block reward.
6. **Proof of Elapsed Time:** PoET is one of the fairest consensus algorithms which chooses the next block using fair means only. It is widely used in permissioned Blockchain networks. In this algorithm, every validator on the network gets a fair chance to create their own block. All the nodes do so by waiting for a random amount of time, adding proof of their wait in the block. The created blocks are broadcasted to the network for others' consideration. The winner is the validator which has the least timer value in the proof part. The block from the winning validator node gets appended to the Blockchain. There are additional checks in the algorithm to stop nodes from always winning the election, and stop nodes from generating the lowest timer value.

What is Ethereum?

Ethereum is a [Blockchain network](#) that introduced a built-in Turing-complete programming language that can be used for creating various decentralized applications(also called Dapps). The Ethereum network is fueled by its own cryptocurrency called 'ether'.

- The Ethereum network is currently famous for allowing the implementation of smart contracts. Smart contracts can be thought of as 'cryptographic bank lockers' which contain certain values.
- These cryptographic lockers can only be unlocked when certain conditions are met.
- Unlike [bitcoin](#), Ethereum is a network that can be applied to various other sectors.
- Ethereum is often called Blockchain 2.0 since it proved the potential of blockchain technology beyond the financial sector.
- The consensus mechanism used in Ethereum is [Proof of Stakes\(PoS\)](#), which is more energy efficient when compared to that used in the Bitcoin network, that is, [Proof of Work\(PoW\)](#). PoS depends on the amount of stake a node holds.

DAO(Decentralized Autonomous Organization) in Blockchain

Last Updated : 23 Jul, 2025



DAO stands for Decentralized Autonomous Organization. The concept of a DAO was first proposed by Bit Shares, Steemit, and EOS (Block. one) founder Dan Larimer in the year 2015, and was further refined in the year 2016 by Ethereum's Vitalik Buterin. A decentralized autonomous organization is decentralized, autonomous, and an organization- as the name already suggests. It is a whole organization that is automated. It stores rules and processes in code. DAOs are often stateless and distributed over millions of computers. No single government could decide to take it down.

Components of DAO:

1. **No central legal entity:** In DAO, there is no central legal entity, this means that no single entity is responsible for regulating the project.
2. **Self-enforcing code:** Smart contracts are created and extensively tested to make sure important details are not overlooked.
3. **Token acts as an incentive for validators:** Tokens are used in DAO for validators to motivate them and to ensure active, fair, and quick participation.

Need of DAO

Beginning an association with somebody that includes financing and cash requires a great deal of confidence in the individuals you're working with. Yet, it's difficult to believe somebody you've just at any point associated with on the web. With DAOs you do not need to trust the other individual within the gathering, simply the DAO's code, which is 100% straightforward and evident by anybody. This opens up countless new freedoms for worldwide joint effort and coordination.

Traditional Organization Vs DAO

S No.	DAO	Traditional Organizations
1.	Casting a ballot is needed by individuals for any progressions to be implemented.	Depending on the structure, changes can be requested from the sole party, or casting a ballot might be advertised.
2.	Votes were counted, and results were carried out consequently without a believed intermediary.	If casting a ballot is permitted, votes are counted inside, and the result of casting a ballot should be taken care of physically.
3.	Completely democratized.	Usually progressive.
4.	Administrations offered are taken care of consequently in a decentralized way.	Requires human taking care of, or halfway controlled mechanization, inclined to control.